

Cooperative Application of Vehicular Traffic Rerouting Method and Adaptive Traffic Signal Control Method

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Abstract— Nowadays, due to the growth in the number of vehicles, the problem of effective traffic control remains relevant. Popular approaches to solve this problem are the use of traffic signal control methods and the use of vehicular rerouting methods. The paper considers the problem of the cooperative application of vehicular traffic rerouting method and the method of adaptive control of traffic signals for effective traffic control. A presented traffic rerouting method takes into account the state of adaptively controlled traffic lights. The results of experimental studies of the developed method in the SUMO simulation environment are presented.

Keywords— intelligent transport system, traffic signal control, vehicular traffic rerouting

I. INTRODUCTION

Currently, the problem of effective control of vehicular traffic remains relevant. Due to the high rate of growth in the number of vehicles on the roads and the relatively low rate of development of transport infrastructure in large cities, the number of traffic jams on the roads is increasing. Traffic congestion, in turn, contributes to increased travel time, increased levels of vehicle fuel consumption, and environmental pollution. According to a report by TomTom, a vehicle navigation systems manufacturer, in 2021 traffic congestion in the city of Samara increased by 3% compared to 2020, and the average loss of time in traffic jams in 2021 was 101 hours [1].

One of the most common ways to reduce the congestion of the transport network is to control the movement of traffic at intersections using traffic lights. In connection with the development of telecommunication technologies and the increase in the number of connected vehicles (CV), adaptive methods for controlling traffic signals have been widely developed. Adaptive control methods use information about the current traffic situation on the traffic lanes adjacent to the intersection to make a decision about choosing the next phase of the traffic light.

In [2], an algorithm for adaptive traffic signals control based on a modification of the well-known Max Pressure algorithm [3] is presented. The developed algorithm uses the vehicle delay time as a metric, unlike the original algorithm, which uses the number of vehicles as such a metric. The results of experimental studies showed the superiority of the developed modification over the original algorithm. The papers [4,5] present methods for controlling the phases of traffic lights based on reinforcement learning. In [6,7], an adaptive traffic signal control method based on a

deterministic predictive model is presented. Within the framework of the developed method, the phase selection was carried out on the basis of the prediction of the weighted vehicle flow crossing the intersection for each phase of the traffic lights on the intersection.

Another approach to improve the efficiency of transport infrastructure is vehicular traffic rerouting depending on the observed traffic situation. This approach allows to reduce the load of congested segments of the road network and evenly distribute it. The authors of [8] presented a vehicular traffic rerouting system based on the use of a neural network to predict the future state of the road network, on the basis of which the route is formed. In [9], a vehicle route rebuilding method and five algorithms for directly constructing a route are presented, using information not only about the congestion of transport network segments, but also information about the routes of each vehicle.

This paper presents a modified version of vehicular traffic rerouting [9] that takes into account information about the state of adaptively controlled traffic lights.

II. ADAPTIVE TRAFFIC SIGNAL CONTROL METHOD

As a method of adaptive control of traffic signals, it is proposed to use the method of maximum weighted flow MaxPWFlow [6,7]. Within the framework of this method, the choice of the next phase of the traffic light is carried out according to the results of the prediction of traffic flow passing through the intersection during the period of the considered phase, for each phase of the working cycle of the traffic light. As a result of the method, the phase for which the predicted traffic flow is maximum is selected.

The detailed description of the adaptive traffic light control method MaxPWFlow is presented below in pseudocode form (Algorithm 1), where τ_{min} is minimum phase switching interval; t_p is the duration of the current active phase $p \in P$ of the traffic signal; P is the set of traffic signal phases, $phase$ is the next phase, selected by MaxPWFlow method.

Algorithm 1: Maximum weighted flow method

Input data: τ_{min} , t_p , p , P

Output data: $phase$

if $t_p < \tau_{min}$ **then**

$t_p = t_p + 1$

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    phase = p
else
    tp = 0
    phase = argmax({PFlow(p) for p in P})
end if

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The predicted "weighted" flow of vehicles is calculated using the following formula:

$$PFlow(p) = \sum_{l \in L_p^{income}} \sum_{c \in C_l} \eta(c, l) I(t(c) < \tau_{min}),$$

where L_{phase}^{income} is the set of incoming lanes to the considered intersection; C_l is the set of vehicles located on the lane l ; $t(c)$ is predicted time based on a deterministic predictive model [6] required by the vehicle c to reach the intersection; I is the indicator function; $\eta(c, l)$ is the vehicle waiting time coefficient [10].

III. VEHICULAR TRAFFIC REROUTING METHOD

The vehicular traffic rerouting method presented in [8] consist of four stages that are performed periodically:

- collecting and preprocessing of vehicles data;
- detection of heavy loaded segments of the road network;
- selection of vehicles to be rerouted and its ranking;
- building a new route for selected vehicles.

Let's take a closer look at each of these steps of the method.

A. Information Collecting and Preprocessing

At the first stage, information about the current state of the road network is collected and preprocessed. Let the transport network is represented as a directed weighted graph G , the edges of which correspond to the road segments of the transport network and the nodes correspond to intersections. It is assumed that the vehicular rerouting system is informed about the position, speed, and route of all vehicles in the transport network. The weight of the transport network graph edge represents the current travel time estimation for the corresponding road segment. The authors of [9] use Greenshield's model [11] to get travel time estimation of a road segment. Due to this model, travel of road segment i is estimated as follows:

$$T_i = \frac{L_i}{V_i^f (1 - \frac{K_i}{K_i^{jam}})},$$

where L_i is the length of road segment i ; V_i^f is the free flow speed of road segment i ; K_i is the current number of vehicles on road segment i ; K_i^{jam} is the maximum possible number of vehicles on road segment i . As was mentioned above the weight of transport network graphs updates

periodically. Based on these weight coefficients of the road network graph, the rerouting process is performed on the last stage of the considered method.

B. Congestion Detection

At the second stage, road segments with high congestion levels are selected. In the original method the following congestion level metric is used:

$$cong_i = \frac{K_i}{K_i^{jam}}.$$

The road segment i is marked as congested if $cong_i > \delta$, where $\delta \in [0, 1]$ is the threshold value. However, the use of such a metric has a significant drawback, which is especially noticeable when using the adaptive method of traffic signal control. It is possible that a road segment has a high value of metric (3), but due to the lack of traffic on the competing road segments of the intersection, this «congestion» does not have a negative effect on traffic flow. Therefore, it is proposed to use the following congestion level metric:

$$cong_i = 1 - \frac{V_i}{V_i^f},$$

where V_i is the average speed of vehicles on road segment i at the previous time step. In experimental studies we empirically set $\delta = 0.1$.

C. Vehicles selection and ranking

Search for nearby vehicles to reroute is performed when there are signs of congestion on a certain road segment. In particular, vehicles located on incoming segments to the considered congested segment are selected for rerouting. The distance parameter L indicates the maximum distance in segments from the congested segment where the farthest vehicle can be located, for which the rerouting will be performed. Then the sets of vehicles selected for rerouting for each congested road segment are combined and ranked.

The effect of a congested road segment on the travel time of a vehicle depends on the remaining distance to the vehicle's destination. It's clear that congestion differently affects the drivers who are close to their destination and drivers who are far from their destination. The considered method uses the urgency metric to rank the vehicles selected for rerouting. Urgency is defined as the difference between the remaining travel time in the current traffic situation and the remaining travel time when driving at the maximum allowed speed. The ranking of the vehicles selected for rerouting is an important step because the route construction algorithm used at the final stage of the method under consideration takes into account information about the route of all vehicles in order to exclude the possibility of new traffic jams. Therefore, vehicles with higher urgency are rerouted first and get relatively better routes.

D. Vehicles Rerouting

In contrast to the original method, the proposed modification of this method presented in this paper takes into account the state of the traffic light-controlled intersections closest to the corresponding vehicle by graph weights adjustment. For each vehicle selected in the previous step of

the method, the road network graph weights are adjusted as follows. The original road network graph is transformed into a graph which edges correspond to road lanes and nodes correspond to connections between lanes. In this case, the weight coefficients of the road segments of the original graph are transferred to the corresponding road lanes in the new graph. If a road segment consists of multiple lanes, then each such lane is assigned a weight coefficient of the original road segment. Connections between lanes in an intersection (internal connections) initially have zero weight. An example of internal lane connections at a traffic light-controlled intersection is shown in Figure 1.

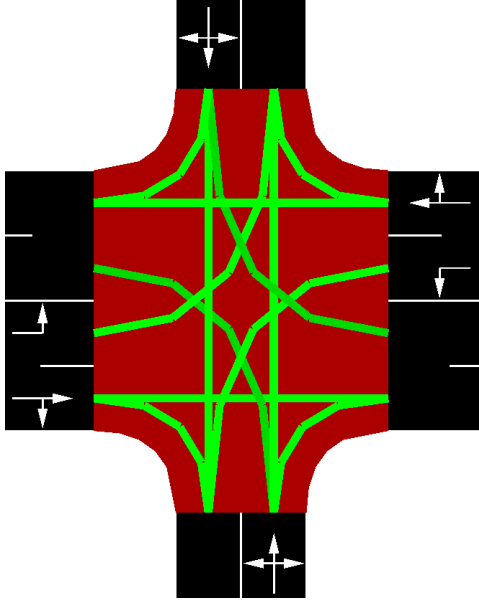


Fig. 1. An example of internal lanes connections (green lines) at an intersection.

Then, separately for each vehicle, the selection of the nearest traffic lights-controlled intersections is carried out, for which the weight coefficients will be updated. The weight coefficients are set at only a limited number of nearby intersections due to the loss of relevance of information about the state of traffic lights over time. Traffic light-controlled intersections are selected that are remote from the considered vehicle by a travel time not exceeding the rerouting time period $\tau_{reroute}$.

Within the proposed vehicular traffic rerouting method modification, the traffic signals are controlled by the adaptive control method described in section II. Let the rerouting time period $\tau_{reroute}$ be equal to the minimum phase switching interval of traffic lights τ_{min} ; τ_{yellow} be a yellow traffic signal time; p_{in}^{out} be the current active traffic signal for connection between incoming lane in and outgoing lane out , $\tilde{p}_{in}^{out}$ be the next traffic signal for connection between incoming lane in and outgoing lane out . Then the internal connection weight of the controlled intersection w_{in}^{out} is calculated as follows:

$$w_{in}^{out} = \begin{cases} 0, p_{in}^{out} = green \wedge \tilde{p}_{in}^{out} = green \\ \tau_{min}, p_{in}^{out} = green \wedge \tilde{p}_{in}^{out} = red \\ \tau_{min} + \tau_{yellow}, p_{in}^{out} = red \wedge \tilde{p}_{in}^{out} = red \\ \tau_{yellow}, p_{in}^{out} = red \wedge \tilde{p}_{in}^{out} = green \end{cases}$$

To build the route of the vehicle, the authors presented a modified version of the well-known search algorithm A* called A* with repulsion (AR*). This algorithm uses information about the current routes of all vehicles in the transport network when calculating the shortest path in order to prevent the formation of new traffic jams in the future. Vehicular traffic rerouting method may cause traffic jams in the future if the constructed vehicles routes contain the same road segment traversed at the same time in the future. Since the proposed method has access to information about the routes of all vehicles in the road network, it is possible to estimate the future impact (footprint) of each vehicle in the road network on its congestion level. A weighted footprint of a road lane i is defined as follows in the original method:

$$f_i = n_i \frac{len_{avg}}{len_i} \frac{V_{favg}}{V_{fi}},$$

where n_i is the total number of vehicles which routes include road lane i ; len_{avg} is the average length of lane in transport network; len_i is the length of road lane i ; V_{favg} is the average free-flow speed of the transport network; V_{fi} is the free-flow speed of road lane i . So, the higher the value of f_i , the less preferable the considered road lane. However, the farther the route segment is located from the initial position of the vehicle, the less it should have an impact on the rerouting process, since rerouting can be repeated while the vehicle is moving to its destination. Therefore, we propose to use the following definition of n_i :

$$n_i = \sum_{c \in C} \sum_{j=0}^{len_{r_c}-1} \alpha^j I(r_{c_j} = i),$$

where C is the set of all vehicles in the transport network; len_{r_c} is the length (in a number of edges) of route r_c of vehicle c ; r_{c_j} is the j -th road lane of route r_c ; α is the coefficient of impact. We empirically set $\alpha = 0.2$.

The heuristic function of AR* algorithm $F(x)$ was changed in comparing with original A* algorithm. It includes the information of vehicles routes as a repulsive force. The repulsive score $R(x)$ of graph node x is defined as the sum of the weighted footprints from the initial node to the node x . So, the path-cost function is defined as follows:

$$F(x) = (1 - \beta) \times (G(x) + H(x)) + \beta \times R(x),$$

where $G(x)$ is the travel time from the initial node to x ; $H(x)$ is heuristic estimation of travel time, which is

calculated as the Euclidian distance between the initial node and node x divided by the maximum allowed speed in transport network; β is the weighting parameter. The parameter β allows a variable weighting between the travel time factor and the repulsive force factor. The value of parameter β was chosen empirically by the authors of original method $\beta = 0.05$.

Based on the above, the one iteration of proposed vehicular rerouting method is presented in a form of pseudocode in Algorithm 2.

Algorithm 2: Vehicular Rerouting Method

Input data: $G, \delta, L, \tau_{reroute}$

Output data: -

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G = updateEdgeWeights(G)
congestedSegments = getCongestedSegments(G,  $\delta$ )
if len(congestedSegments) < 0 then:
    return
end if
vehsToReroute = []
for segment in congestedSegments:
    segVehs = getVehiclesToReroute(segment, L)
    vehsToReroute.append(segVehs)
end for
vehsToReroute = sortByUrgency(vehsToReroute)

G_lane = getLaneBasedGraph(G)
for veh in vehsToReroute:
    fp = getEdgesFootprint()
    current, dest = getVehicleRoute(veh)
    G_lane = updateVehWeights (G_lane, veh,
 $\tau_{reroute}$ )
    newRoute = AstarRepulsion(G_lane, current,
dest, fp)
    setRoute(veh, newRoute)
return

```

IV. EXPERIMENTS

Experimental studies of the effectiveness of the proposed method were carried out using the SUMO (Simulation of Urban Mobility) modeling system [12]. The experiments compared the proposed modified method of traffic rerouting and the original method. At the same time, traffic lights at intersections were controlled by an adaptive traffic signal control method MaxPWFlow.

Two simulation scenarios were used to evaluate the effectiveness of the proposed method:

- The "grid4x4" scenario. A synthetic grid network consisting of 16 intersections of the same structure. The total number of vehicles is 2270.
- The "cologne8" scenario. The transport network is part of the real transport network of the city of Cologne, contains 8 regulated intersections of various structures. The total number of vehicles is 1740.

The road networks of the scenarios under consideration are shown in Figure 2.

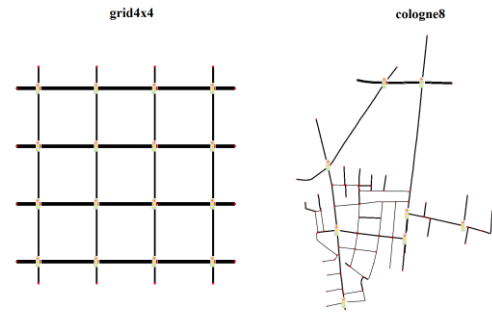


Fig. 2. Road networks of simulation scenarios.

A comparison of vehicular rerouting methods was carried out according to three criteria:

- average fuel consumption of vehicles;
- average travel time of vehicles;
- average waiting time of vehicles.

All vehicles in each of the simulation scenarios have the same characteristics, including the same Euro 4 gasoline engines. During the experiments, the simulation was performed 10 times for each of the considered scenarios, the results were averaged. The starting positions and departure times of the vehicles were different in different episodes of the same scenario. Experimental studies were carried out with a simulation step of 1 second and a total simulation time of 3600 seconds. The results of comparing the methods by the criterion «average waiting time» are presented in Table I, by the criterion «average travel time» - in Table II, and by the criterion «fuel consumption» - in Table III.

TABLE I. COMPARISON OF METHODS BY «AVERAGE WAITING TIME» CRITERION, S

Scenario	Proposed method	Original method	Without rerouting
«grid4x4»	38.59	42.97	192.40
«cologne8»	3.49	3.89	3.92

TABLE II. COMPARISON OF METHODS BY «AVERAGE TRAVEL TIME» CRITERION, S

Scenario	Proposed method	Original method	Without rerouting
«grid4x4»	188.73	202.07	397.90
«cologne8»	84.31	89.21	89.28

TABLE III. COMPARISON OF METHODS BY «AVERAGE FUEL CONSUMPTION» CRITERION, ML

Scenario	Proposed method	Original method	Without rerouting
«grid4x4»	200516.06	210352.41	356808.55
«cologne8»	70391.32	74023.71	74050.43

The conducted experiments confirm the effectiveness of the proposed vehicular traffic rerouting method in cooperation with the adaptive traffic signals control method according to all the considered criteria.

V. CONCLUSION

In this paper, we presented a modification of the vehicular traffic control method, which takes into account the information about the state of traffic lights controlled by the adaptive method. The experimental analysis of the proposed method was carried out in the traffic simulation environment SUMO. The obtained results show that the cooperative application of the proposed vehicular rerouting method and adaptive traffic control method can improve the effectiveness of the transport network.

Further research will be focused on the analysis of the cooperative application of vehicular rerouting methods, vehicle trajectory planning algorithms, and adaptive traffic light control methods.

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